

SAKTHI MAHESHWARI RAMKUMAR

Computer Science Undergraduate | +65 96561495 | rsrsakthi7@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Computer Science undergraduate with experience building full-stack applications in Python, JavaScript, and SQL, alongside machine learning and data science projects. Comfortable working across the stack, from backend logic and database operations to testing and debugging, and have applied OOP and exception handling through coursework and personal projects. Motivated to apply these skills in a software engineering and AI-focused environment.

EDUCATION

Singapore Institute of Management – University of London

Singapore

Bachelor of Science (Honors) in Computer Science

Expected Sep 2026

- **Relevant Coursework:** AI & ML, OOP, Algorithms & Data Structures, Computer Security, NLP, Data Science, Discrete & Computational Mathematics, Software Design & Development

WORK EXPERIENCE

HA Academy

Singapore

Associate Trainer (Flexible/Ad hoc)

Oct 2024 – Present

- Facilitated workshops for 100+ students across schools, developing communication and presentation skills. Mentored students during camps, demonstrating adaptability and leadership in fast-paced environments.

PROJECTS

FYP Multimodal RAG-Based Learning Assessment System – Conversational AI, RAG, LLMs

May 2026 – Present

- Developing a multi-model AI learning assessment system to evaluate voice-based explanations and generate feedback.
- Orchestrating speech-to-text, retrieval-augmented generation (RAG), and language model components into a single pipeline.
- Designing data processing pipelines and workflow orchestration across multiple AI components.

Event Management Web Application – SQL, Node.js, Express.js, EJS, HTML/CSS/JS

Jul 2025

- Built a full-stack event ticketing app with Node.js/Express.js backend, EJS frontend, and SQLite database.
- Designed a relational schema with 5 linked tables and 10+ RESTful API endpoints with full CRUD operations.
- Applied server-side validation and exception handling to prevent overbooking and invalid bookings.

Crypto CLI Trading Simulator – C++, OOP, Data Structures

Jan 2025

- Built a CLI trading simulator using OOP, with modular classes for authentication, wallet, and transaction processing.
- Processed historical trading data via file I/O to compute OHLC candlestick summaries across timeframes.
- Implemented password hashing, access control, and unit testing across transactions and edge cases.

Brain Tumour MRI Classification – Python, PyTorch, CNNs

Jan 2026

- Built a pipeline using YOLOv8 for localisation and ResNet-18/EfficientNet-B0 for classification, hitting 96.9% accuracy.
- Designed a preprocessing pipeline for 4,000+ MRI images using OpenCV to improve model generalisation.
- Applied semi-supervised pseudo-labelling to scale localisation from 250 labelled samples to the full dataset.

LEADERSHIP & ACTIVITIES

Data Analytics Club – Singapore Institute of Management

Sep 2025 – Mar 2026

- Worked on internal data science projects with peers, improving understanding of data analysis concepts and strengthening interest in AI and applied analytics.

SIM-UOL Computer Science Student Council Hackathon 2024

Nov 2024

- Collaborated in a 5-member team to design and develop a browser-based scrolling game within a 2-day hackathon, focusing on rapid development, debugging and task coordination under tight time constraints.

Basketball Team Captain – Bedok View Secondary School

Jan 2019 – Dec 2020

- Led the school basketball team during competitions, coordinating training, teamwork, and on-court communication.

Volunteer Activities – Team Nila, Youth Corps Singapore, SCS

2021 – Present

- Participated in community initiatives including digital assistance for elderly users (e.g., SimplyGo), sports and food distribution, and environmental activities, engaging with diverse groups across service-oriented projects.

SKILLS & CERTIFICATIONS

Programming Languages: Python, JavaScript, C++, SQL

Software & Web Development: Node.js, Express.js, HTML, CSS, REST APIs, Object-Oriented Programming

Databases: MySQL, SQLite

AI, Machine Learning & Data Science: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, RAG, LangChain

Tools & Platforms: Git, GitHub, GitLab (Basic Understanding), Jupyter Notebook, Google Colab

Certifications: [AWS Cloud Foundations](#)